

1 (a) $v_x = (6.0^2 - 4.8^2)^{1/2}$ M1

$= 3.6 \text{ m s}^{-1}$ A0

or

$6.0 \sin \theta = 4.8$, so $\theta = 53.1^\circ$ and $v_x = 6.0 \cos 53.1^\circ$ M1

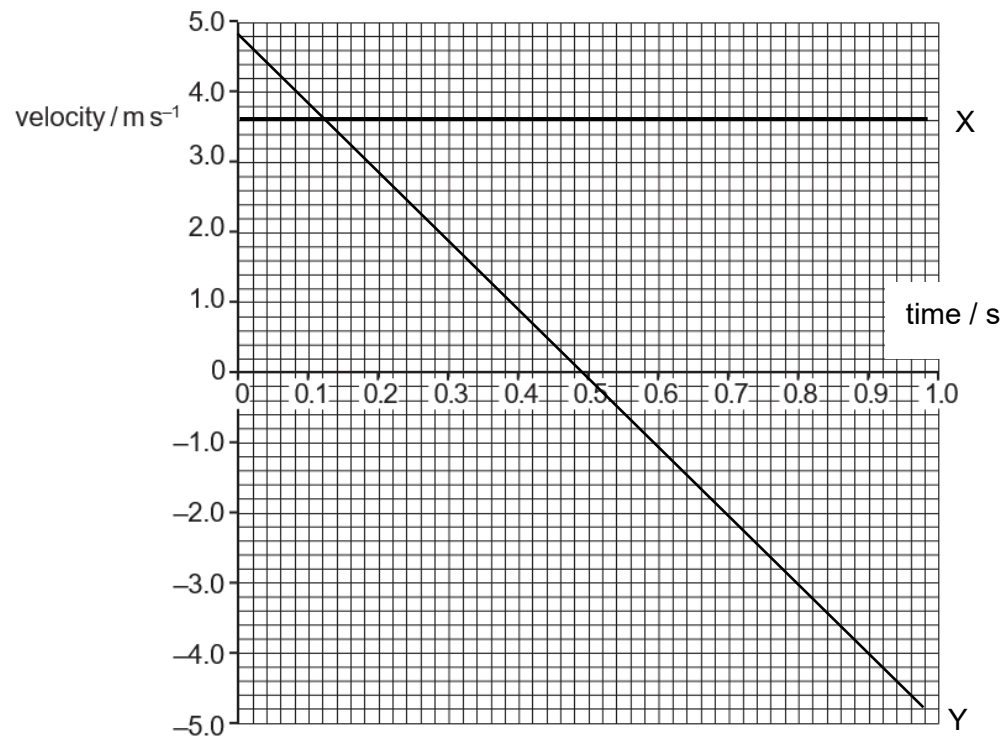
$= 3.6 \text{ m s}^{-1}$ A0

(b) (i) straight line from (0, 4.8) to (0.49, 0) A1

straight line continues with same slope to (0.98, -4.8), labelled Y A1

(ii) a horizontal line A1

from (0, 3.6) to (0.98, 3.6), labelled X A1



(c) $s = ut + \frac{1}{2}at^2 = (4.8 \times 0.49) + (\frac{1}{2} \times -9.81 \times 0.49^2)$ C1
 $= 1.2 \text{ m}$ A1

or $s = \frac{1}{2}(u + v)t = \frac{1}{2} \times (4.8 + 0) \times 0.49$ C1
 $= 1.2 \text{ m}$ A1

or $v^2 = u^2 + 2as$
 $s = 4.8^2 / (2 \times 9.81)$ C1
 $= 1.2 \text{ m}$ A1

2

(d)
$$\frac{\text{kinetic energy at maximum height}}{\text{change in gravitational potential energy}} = \frac{\frac{1}{2}(mv_x^2)}{mgh} = \frac{\frac{1}{2}(3.6^2)}{9.81(1.2)} \quad \text{C1}$$
$$= 0.56 \quad \text{A1}$$

or

$$\frac{\text{kinetic energy at maximum height}}{\text{change in gravitational potential energy}} = \frac{\frac{1}{2}(mv_x^2)}{\frac{1}{2}(mv_Y^2)} = \frac{\frac{1}{2}(3.6^2)}{\frac{1}{2}(4.8^2)} \quad \text{C1}$$
$$= 0.56 \quad \text{A1}$$

- (e) With air resistance, the resultant force is larger than weight, resulting in larger deceleration, hence the actual time is shorter. **B1**

3 (a) energy (stored) / work done represented by area under graph **B1**